

Intro to R — Stat 5810, Section 001, Spring 2019 (1 Credit)

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Office Hours: Monday (M), Wednesday (W) & Friday (F), 1/7/19 – 2/11/19, 2:30pm – 3:30pm and by appointment; Tuesday (T) & Thursday (H), 2/12/19 – 4/11/19, 1:30pm – 2:30pm and by appointment.

Classes & Rooms:

MWF 1:30pm – 2:20pm, M 1/7/19 – M 2/11/19 (tentatively): ENGR 205.

Please visit the course Web page listed above and/or Canvas frequently for lecture notes, data sets, R code, etc. — in particular if you miss class for any reason.

Detailed Class Schedule:

For a 1-credit course, we need 15 lectures/lecture days (in contrast to 43 or 44 lectures/lecture days for a 3-credit course). Those days are marked as “Lecture 01” to “Lecture 15” in the overview below:

Week	Monday	Wednesday	Friday
1	1/7: Lecture 01	1/9: Lecture 02	1/11: Lecture 03
2	1/14: Lecture 04	1/16: Lecture 05	1/18: Lecture 06
3	1/21: MLK Day	1/23: Lecture 07	1/25: Lecture 08
4	1/28: Lecture 09	1/30: Lecture 10	2/1: Lecture 11
5	2/4: Lecture 12	2/6: Lecture 13	2/8: Lecture 14
6	2/11: Lecture 15	2/13: Backup	2/15: Backup

Note: I have marked a few days as “Backup”, e.g., in case we miss lectures because of snow days, I am sick, or I have to travel. But, hopefully, this won’t happen. If nothing goes wrong, our tentative last lecture date will be on M 2/11/19.

Course Objectives:

The purpose of this course is to provide students from statistics and other departments with a sound foundation of the R software environment. The course is not designed to teach students to use R for data analysis or visualizations, but to prepare students to use R *effectively* in other statistics courses.

Students will learn how to read, write, and understand base R code. They will learn about R’s data structures, function language, and documentation. They will learn simple programming skills, including how to write their own functions. At the end of the course, a student should be able to “use” an R function and extract the results, read and write data files, and produce an R Markdown document combining text and analysis.

Prerequisites:

I do not expect any prior knowledge of R. Knowledge from an introductory statistics course such as Stat 2000, Stat 3000, or higher is required (Stat 1040 or Stat 1045 is not enough). You should, for example, know basic probability distributions (Bernoulli, Binomial, Gaussian) and know what the Central Limit Theorem (CLT) says.

No programming experience is required, although you must be familiar with your preferred operating system (Windows, Mac, or Linux) and understand how to download files from the web, manage folders, install software, etc.

IDEA Center Learning Objectives:

Objective 1) Gaining factual knowledge (terminology, classifications, methods, trends).

Objective 3) Learning to apply course material (to improve thinking, problem solving, and decisions).

Objective 4) Developing specific skills, competencies, and points of view needed by professionals in the field most closely related to this course.

Topics: (subject to change)

1. Introduction
2. Getting started with R, RStudio, and R Markdown
 - (a) Installing and running R, RStudio, and R Markdown
 - (b) Using R Markdown to write reports (and more!)
 - (c) Finding functions, objects and documentation
 - (d) Issuing commands and saving results
3. The R Language
 - (a) Data types
 - (b) Subsetting
 - (c) Vectors
 - (d) Matrices
 - (e) Data frames
 - (f) Lists
 - (g) Apply
 - (h) Functions
4. Programming
 - (a) Control flow: if, for, while
 - (b) Vectorization and efficiency
 - (c) Writing your own functions

Note that this course does not introduce any methods how to visualize and summarize statistical data and results via statistical graphics (such as bar charts, histograms, dot plots, box plots, etc.). To learn how to do so, you should attend “Statistical Visualization I” that usually is offered in the Fall.

Homework Assignments:

There will be about 5 ± 1 HW assignments for our course, roughly one per week. Each assignment will include a value (typically 20–100 points) that it will be scored out of. Your

final grade will be determined by the sum of your points in all assignments. The value of each assignment will be roughly proportional to its importance and the amount of work involved.

You will be allowed to discuss general approaches to questions on the assignments with other students, but each student must write and submit their own code and comments. Any students caught sharing code will fail the class.

Unless otherwise stated on the assignment sheet, all homework assignments have to be submitted electronically via Canvas.

The following deductions will be applied to late homework submissions: 1 min – 24 hours late: 10% off; > 24 hours – 48 hours late: 25% off; > 48 hours – 72 hours late: 50% off. Homeworks won't be accepted later than 72 hours (i.e., 3 days) after the submission deadline.

There will be no (in-class or take-home) quizzes, midterm exams, or final exams. We will have a few in-class worksheets for training purposes only.

Textbooks:

There is no formal textbook for this class, but the instructor will provide links to suitable online resources.

Software:

We will primarily be using R (<http://cran.r-project.org/>), a free software environment for statistical computing and graphics. Please install a recent version of R, i.e., 3.5.1 or 3.5.2, on your own computer so we can exchange code. Also install RStudio (<https://www.rstudio.com/>) as a front end to R and MiKTeX (<https://miktex.org/>) that will allow us to combine code and results from R into text documents. We will jointly look into the software installation in our first two lectures.

Courtesy:

Please turn off cell phones and similar devices before class, and please keep conversations to a minimum during lectures. Please do not read/reply to your e-mails or browse other web pages than the ones discussed during class.

I will not keep track if you come to class or not. However, I would highly recommend to attend all lectures. If you have to miss a lecture, there will be a recording of the lecture available in Canvas Panopto (if the technology doesn't fail).

Americans with Disabilities Act:

If a student has a disability that will likely require some accommodation by the instructor, the student must contact the instructor and document the disability through the Disability Resource Center (DRC), preferably during the first week of the course. Any requests for special considerations relating to attendance, pedagogy, taking of examination, etc. must be discussed with and approved by the instructor. In cooperation with the Disability Resource Center, course materials can be provided in alternative formats — large print, audio, or Braille.

Note:

The above schedule and procedures in this course are subject to change in the event of extenuating circumstances.